Lorenzo Orrante

+52 33 2275 1129 lorenzo.orrante@icloud.com linkedin.com/in/lorenzoorrante github.com/ZeusTucZ

Education

Tecnológico de Monterrey | GPA: 3.75/4.00 Aug 2024 – Expected June 2028
B.S. in Computer Science and Technologies *Monterrey, Nuevo León*

Engineering Experience

VantTec Autonomous Systems Team Sep 2026 – Present
Tecnológico de Monterrey *Monterrey, Nuevo León*

- Developed an **autonomous drone navigation system** in Python using OpenCV and djitellopy to detect and align with a 1×1 m visual gate.
- Debugged and iterated on a **computer-vision feedback loop**, reproducing alignment failures and continuously correcting drone movement from camera-frame error during takeoff, traversal, and landing.

Elyos Racing Team – **[6th Place – Shell Eco-marathon Americas 2025]** Aug 2025 – Jun 2026
Tecnológico de Monterrey *Guadalajara, Jalisco*

- Led race strategy for Shell Eco-marathon Americas 2025, building a **Python energy model** to evaluate lap strategies and engineering data, achieving an **11% improvement in energy efficiency**.
- Developed a **physics-based race simulator** in Python and collaborated with mechanical and electronics sub-teams to validate engineering assumptions and support data-driven decisions.

Projects

Vehicle Telemetry Dashboard | *React, Express.js, PostgreSQL, Socket.io* May 2025 – Apr 2026

- Engineered a **real-time telemetry and monitoring platform** using React, Express.js, PostgreSQL, and WebSockets, achieving **<50ms communication latency** across 100 validated data points.
- Implemented **automated data validation** and backend persistence for incoming telemetry, improving reliability of data used for real-time and historical analysis.
- Developed and **deployed** the platform to a cloud environment, supporting reliable monitoring during track testing and helping identify abnormal or inconsistent system behavior.
- Built a responsive React interface for live metrics and historical data, enabling faster inspection of vehicle performance during testing sessions.

Lux: Local AI Meeting Assistant | *Python, FastAPI, Ollama, Whisper, ChromaDB, React* May 2026

- Built an end-to-end **Python/FastAPI application** that transcribes audio with Whisper and generates summaries, mind maps, and chat responses with a local Ollama LLM.
- Designed a **RAG pipeline** with ChromaDB that reduced input tokens by **30%** by retrieving only relevant transcript context.
- Integrated backend services with a React interface, implementing streaming responses, persistent chat history, timeline navigation, and structured transcript access.

Smart Mirror – [1st Place – Expo Ingenierías] | *Python, JavaScript, MongoDB* Oct 2025 – Dec 2025

- Built a **Smart Mirror** with facial recognition and **on-device inference**, keeping biometric data within the local network through a Privacy-by-Design approach.
- Developed and integrated **voice command functionality** for hands-free interaction with the application.
- Awarded **1st place** in the Physical Prototype category at Expo Ingenierías, competing against 12+ engineering teams.

Technical Skills

Programming: Python, C++, JavaScript/TypeScript, MATLAB
Testing & Validation: Debugging, Automated Validation, API Testing, Git, GitHub
Systems & Infrastructure: Linux, Cloud Deployment, WebSockets, PostgreSQL, MongoDB
Software Engineering: Object-Oriented Programming, Data Structures & Algorithms, REST APIs, Full-Stack Development, FastAPI, Express.js, React
AI & Computer Vision: OpenCV, Ollama, ChromaDB, Whisper
Certifications: OCI Foundations Associate, OCI AI Foundations Associate, Data Structures (ZTM)